

Algorithms and Data Structures Lab 03

Question 01)

```
stack1 = []

stack1.append(20)
print (stack1)
stack1.append(30)
print (stack1)
stack1.append(40)
print (stack1)

stack1.append(50)
print (stack1)

stack1.pop()

print("new stack is", stack1)

print("top element ", stack1[-1])
```

Output –

```
[20]
[20, 30]
[20, 30, 40]
[20, 30, 40, 50]
new stack is [20, 30, 40]
top element  40
```

Question 02)

```
stack2 = []

no = int(input("Enter the number of Elements: "))

for i in range(no):
    element = int(input("Enter the elements: "))
    stack2.append(element)

print("Stack is", stack2)

if len(stack2) > 0:
    stack2.pop()
    print("Updated stack is", stack2)

    if len(stack2) > 0:
        print("Top element is", stack2[-1])
    else:
        print("Stack is empty")
else:
    print("Stack is already empty")
```

Output –

```
Enter the number of Elements: 5
Enter the elements: 2
Enter the elements: 4
Enter the elements: 6
Enter the elements: 8
Enter the elements: 10
Stack is [2, 4, 6, 8, 10]
Updated stack is [2, 4, 6, 8]
Top element is 8
```

Question 03)

```
stack = []

stack.append("A")
stack.append("B")
stack.append("C")
stack.append("D")

print("Stack is " , stack)

stack.pop()
stack.pop()

print("Remaining characters", stack)
```

Output –

```
Stack is  ['A', 'B', 'C', 'D']
Remaining characters ['A', 'B']
```

Question 04)

```
queue= []

queue.append(10)
print(queue)

queue.append(20)
print(queue)

queue.append(30)
print(queue)

queue.append(40)
print(queue)

queue.append(50)
print(queue)

queue.pop

print("Updated queue:", queue)
print("Front element:", queue[0])
```

Output –

```
[10]
[10, 20]
[10, 20, 30]
[10, 20, 30, 40]
[10, 20, 30, 40, 50]
Updated queue: [10, 20, 30, 40, 50]
Front element: 10
```

Question 05)

```
queue = []

number = int(input("Enter the number of elements: "))

for i in range(number):
    value = int(input("Enter an element: "))
    queue.append(value)

print("Queue: ", queue)

removed = queue.pop(0)
print("Removed element: ", removed)
print("Updated queue: ", queue)

if len(queue) > 0:
    print("Front element: ", queue[0])
else:
    print("Queue is empty")
```

Output –

```
=====
Enter the number of elements: 3
Enter an element: 2
Enter an element: 4
Enter an element: 6
Queue: [2, 4, 6]
Removed element: 2
Updated queue: [4, 6]
Front element: 4
|
```

Question 07)

```
stack = []
queue = []

for i in range(5):
    element = int(input("Enter a value: "))
    stack.append(element)
    queue.append(element)

print("Stack", stack)
print("Queue", queue)

stackremove = stack.pop()
queueremove = queue.pop(0)

print("Removied from stack", stackremove)
print("Removied from queue", queueremove)

print("Stack", stack)
print("Queue", queue)
```

Output –

```
=====
Enter a value: 20
Enter a value: 30
Enter a value: 40
Enter a value: 50
Enter a value: 60
Stack [20, 30, 40, 50, 60]
Queue [20, 30, 40, 50, 60]
Removied from stack 60
Removied from queue 20
Stack [20, 30, 40, 50]
Queue [30, 40, 50, 60]
```